

Experiment No. 8

Experiment Name

Customer Churn Prediction using Decision Tree Classification

Aim

To build and evaluate a Decision Tree Classification model for predicting customer churn using a real-world dataset and analyze the factors influencing customer retention.

Purpose

Customer churn prediction is a critical application of supervised machine learning that helps organizations identify customers who are likely to discontinue a service or product. Early identification of such customers enables businesses to implement targeted retention strategies, improve customer satisfaction, and reduce revenue loss.

In this experiment, students will use datasets such as the Telco Customer Churn Dataset, Bank Customer Churn Dataset, or IBM HR Employee Attrition Dataset to develop a Decision Tree classifier, evaluate its performance, and interpret the generated decision rules.

Prerequisites

- Basic knowledge of Python programming
- Familiarity with Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn
- Understanding of supervised machine learning and classification concepts
- Knowledge of data preprocessing and feature encoding
- Jupyter Notebook or Google Colab

Steps

1. Select a suitable dataset such as the Telco Customer Churn Dataset, Bank Customer Churn Dataset, or IBM HR Employee Attrition Dataset.
2. Import the required Python libraries and load the dataset into a Pandas DataFrame.
3. Perform data preprocessing by handling missing values, encoding categorical variables, and selecting relevant features for prediction.
4. Split the dataset into training and testing sets using an appropriate train-test ratio (e.g., 80:20).
5. Train a Decision Tree Classifier using the training dataset and configure suitable parameters such as maximum tree depth and splitting criteria (Gini Index or Entropy).

6. Use the trained model to predict customer churn for the testing dataset.
7. Evaluate the model using a Confusion Matrix, Accuracy, Precision, Recall, and F1-Score.
8. Visualize the Decision Tree to understand the decision-making process and identify the most influential features.
9. Analyze the classification results and determine the key factors contributing to customer churn.
10. Summarize the findings and recommend business strategies for improving customer retention based on the model's predictions.

Questions

1. What is classification, and how does it differ from regression?
2. Explain the working principle of the Decision Tree algorithm.
3. Differentiate between Gini Index and Entropy as splitting criteria.
4. What is a Confusion Matrix? Explain its components.
5. Define Accuracy, Precision, Recall, and F1-Score. Why are these metrics important?
6. What is overfitting in a Decision Tree? How can it be reduced?
7. Why is customer churn prediction important for businesses?
8. What are the advantages and limitations of the Decision Tree algorithm?
9. Mention three real-world applications of Decision Tree Classification other than customer churn prediction.
10. How can the insights obtained from a churn prediction model help organizations improve customer retention and business profitability?